



DATA ANALYTICS

Session 3: Prepare Data

Agenda

- Session 1: Setup
- Session 2: Explore Data
- **Session 3: Transform Data**
- Session 4: Share Insights

EXPLORE DATA

Outline

- Inspect
- Subset
- Edit
- Summarize
- Visualize

Data Exploration

- Data exploration is a crucial first step in any machine learning workflow.
- Helps data scientists understand the
 - *structure,*
 - *quality, and*
 - *patterns in the data.*

Inspect Data

- Examine data format
 - *Plain text (e.g., csv, .txt), spreadsheet (e.g., .xlsx), columnar/binary (e.g., parquet, feather, pickle), or statistical format (e.g., SPSS, SAS)*
- Examine data to gain insight on
 - *size*
 - *structure*
 - *data types*
 - *completeness*
 - *consistency*
 - *encoding*

Subset

Machine learning problems often involve

- selecting relevant features
- using observations that meet analysis requirements
 - *For e.g.: individuals who have visited the website in the last one month, a random sample of 70% of data*

Edit Data

- This is essential for making datasets cleaner, more interpretable, and analysis-ready.
- Change variable names to align with analysis needs.
 - *For instance, names that are too long, uninformative or do not comply with naming rules*
- Derive new variables from existing variables
- Identify missing values and replace them with an identifier or a value
- Change data types
 - *If data was parsed incorrectly; accommodate greater precision; reduce memory requirement*
- Sort
- Delete rows or columns

Summarize

- Eyeballing even a moderate sized dataset is unlikely to yield useful insights
- Numerical summaries can yield insights independent of the actual size of the data
- For numeric variables
 - *Measures of central tendency: Mean, Median, Mode*
 - *Measures of dispersion: Standard Deviation, Variance, Range*
 - *Shape of the distribution: Skewness, Kurtosis*
- For categorical variables
 - *Frequency or Proportion*
 - *Cross-tabulations*

Visualize

- Early in the analysis pipeline, visualizing data can aid in
 - *forming an understanding of the data,*
 - *identifying trends, and*
 - *spotting anomalies*
- Common chart types
 - *Histogram: Distribution of variable*
 - *Scatterplot: Relationship between a pair of variables*
 - *Line plot: Trend over time*
 - *Bar chart: Compare a numeric variable across levels of a categorical variable*

TIDY DATA

Data Preparation

- Data seldom comes in a form that is ready for analysis. This may be because
 - *Variables are not in the correct format*
 - *Variables are too raw for analysis*
 - *Structure of data is not amenable for analysis*
 - *Data is spread across multiple tables*
 - *Some of the data is missing*

Tidy Data

- Parsing: When variables are not the correct format
- Feature Extraction: When variables are too raw for analysis
- Restructuring: When the structure of data is not amenable for analysis
- Joining tables: When data is spread across multiple tables
- Imputation and Deletion: When data is missing

Parsing

- Data imported must assign the correct class to variables:
 - *Numbers using a type like int32 or float64*
 - *Text as string type*
 - *Boolean as Boolean type*
 - *Dates using a suitable date type*
 - *Categorical variables as type category*

Parsing

■ Numbers

- [Conventions for writing numbers](#) differ across the world. For e.g., in the US, we use a period for a decimal point and a comma for grouping numbers but France does the exact opposite.
- Numbers may include characters or words to show context. E.g., \$25, 25%, 25% increase in wages

Parsing

■ Words

- *As such, words are easy to parse.*
- *But this simple task is complicated by the use of special symbols (e.g., Häagen-Dazs), non-english languages, and emojis(e.g., 🍌).*
- *The ASCII system of encoding characters was not designed for such use cases.*
- *This has given way to UTF-8, a more expansive and universal character coding format that extends the lexicon of ASCII.*

Parsing

- Dates

- *The conventions for expressing date and time vary across the world.. For e.g., US Independence Day can be written as*
 - July 4, 1776
 - 4/7/1776
 - 7/4/1776
 - 1776/7/4

Feature Engineering

- Variables contained in a dataset seldom come in an analysis-ready format.
- With suitable transformations, variables can be converted into useful features.
- The nature and types of transformation depend on whether the variable is categorical or numeric.
- Examining all variable transformations is beyond the scope of this course, but we will examine commonly used transformations.

Feature Engineering For Categorical Variables

- **Dummy Variables:** A categorical variable with k levels is represented using $k-1$ dummy variables, where one of the levels serves as a reference level. A dummy variable can only take on two values, 0 and 1. An alternative, one-hot encoding represents k levels with k dummy variables.
- **Remove Low Variance Variables:** Remove variables with zero variance or low variance.
- **Lump categories:** Combine sparse categories with a related category
- **Other Category:** Combine rare categories into an Other category
- **Date and Time:** Elements of date and time such as year and hour
- **Text:** Elements of string type

Feature Engineering For Numeric Variables

- Scaling: Place all predictors in the same units by methods such as centering, standardizing and range scaling.
- Skewness: Fix distribution of skewed variables by applying suitable methods (e.g., Box-Cox, Yeo-Johnson)
- Low Variance: Remove variables with zero variance or low variance.
- Simple Transformations: Transform variables to make it a more predictive feature (e.g., square, cube, interaction)
- Combine Variables: Construct a weighted or unweighted index by combining variables.
- Binning: Group a numeric variable into a set of two more discrete bins.

Restructure Data

- Sometimes, structure of the data may get in the way of analyses
- Hadley Wickham recommends the use of a *tidy* data structure which is one in which
 - *Each variable has its own column*
 - *Each observation has its own row*
 - *Each values has its own cell*
- Choice of structure is often driven by the functions to be used and analytical technique to be conducted.

Restructure Data

Wide Data

```
##      Country  1960      2017
## 1      USA 543300 19390604
## 2    China  59716 12237700
## 3    Japan  44307  4872136
```

Tall Data

```
##      Country Year      GDP
## 1      USA 1960   543300
## 2    China 1960    59716
## 3    Japan 1960    44307
## 4      USA 2017 19390604
## 5    China 2017 12237700
## 6    Japan 2017  4872136
```

Joins

- Data is generally spread across multiple tables rather than in a single flat file.
- These tables share a common variable or field, thus are related (which is why these tables are also known as relations). All the tables or relations together comprise a relational database.
- Storing data across relations offers a number of benefits such as reduced redundancy, fewer errors from updates, and greater efficiency in storage and retrieval.
- The data required for analysis at hand is retrieved by linking relations through joins. The languages most widely used for this is SQL, however, this can also be accomplished using other languages like Python and R.

Joins

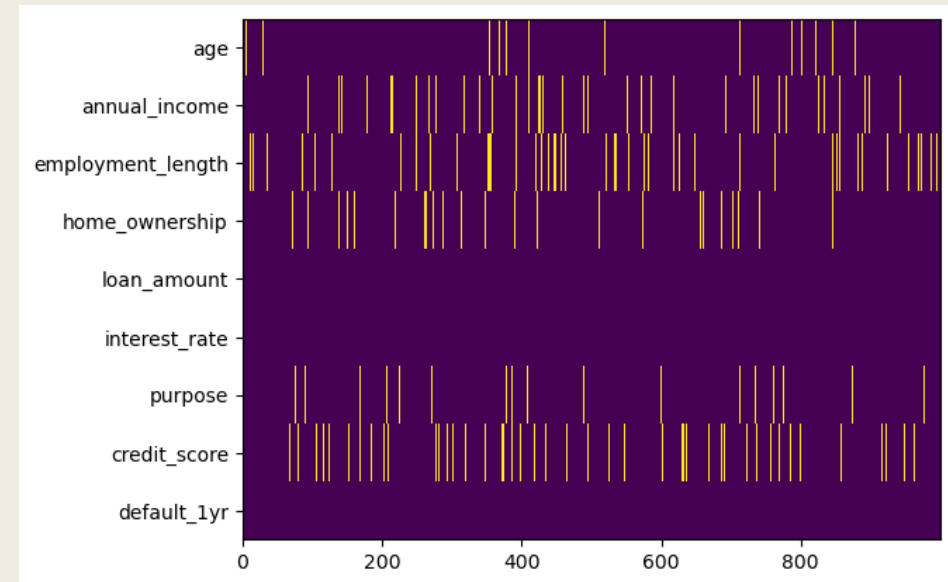
- Types of Joins
 - *Inner Join*
 - *Outer Joins*
 - Left Outer Join
 - Right Outer Join
 - Full Outer Join

Missing Data

- Common in real datasets
- Critical to address it as many predictive models require complete data
- Important to understand reasons for missing data in order to decide what do with it

Missing Data

- Possible reasons may be
 - *Random*
 - *Data deficiency*
 - *Specific causes*
- Pattern in a heatmap may highlight reasons



Missing Data Solutions

1. Delete
2. Encode Missing Data
3. Impute

- Simple solution
- Delete
 - *Pairwise*
 - *Listwise*
- However,
 - *small number of missing values can lead to losing a large percent of sample.*
 - *acquiring new data may be costly*
 - *potential of introducing bias*

Missing Data Solutions

1. Delete
2. Encode Missing Data
3. Impute

- Hard code missingness
- In a categorical variable, missing data may be encoded as another level
- Missing data in a continuous variable may be represented using a dummy variable.

Missing Data Solutions

1. Delete
2. Encode Missing Data
3. Impute

- Information and relationships among non-missing predictors is used to estimate missing values
- Univariate techniques: Use a non-biasing estimate like mean or median.
- Multivariate techniques: Use data on other variables to estimate missing values. E.g, kNN, MICE.
- Impute no more than 20% missing data for any variable
- Conducted prior to other steps in data preparation

PROJECT

Project

- Form a Group
- Select Data
- Explore and Transform Data
- Share Insights